

Customer Support Use Case Worksheet

Instructions: Record your prompts and analysis below. Compare the initial Use Case to the Generative AI revisions.

What are Generative AI recommendations?

ChatGpt:

Generative AI Recommendations for the Customer Support Use Case

1. Introduce an AI-Powered Virtual Assistant

- Deploy a conversational AI chatbot capable of understanding natural-language queries (not limited to menu options).
- Provide 24/7 self-service for common tasks such as:
 - reporting a lost card
 - resetting passwords
 - checking account status
 - receiving fraud alerts
- Reduce call volume by handling routine inquiries automatically.

2. Intelligent Routing & Prioritization

- Use generative AI to classify the customer's issue in real time based on their typed or spoken description.
- Predict urgency (e.g., a lost card = high-priority) and route customers to the right agent or department instantly.
- Reduce wait times by matching expert agents to complex issues.

3. AI-Assisted Agent Support

- Provide agents with real-time suggested responses tailored to the customer's history and issue.
- Enable AI to summarize long interactions so agents avoid repetitive review.
- Offer dynamic troubleshooting steps instead of rigid scripts, improving personalization and first-call resolution.

4. Automated Case Management & Follow-Up

- Automatically generate structured case notes after each interaction.

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- Auto-draft follow-up emails or messages summarizing next steps.
- Trigger automated reminders to ensure unresolved issues are addressed faster than the current 24-hour callback window.

5. Predictive Analytics for Customer Satisfaction

- Use generative AI to identify patterns in complaint topics, wait times, or repeated issues.
- Predict churn risk (customers likely to leave the bank).
- Recommend proactive outreach—e.g., notifying customers about new self-service tools or fraud prevention tips.

6. Personalized Customer Experience

- Tailor support based on:
 - account type
 - transaction patterns
 - past problems
 - customer preferences
- Provide tailored product recommendations or solutions, improving satisfaction and long-term retention.

7. Multi-Channel Integration

- Deploy the AI assistant across:
 - mobile app
 - website
 - IVR phone system
 - SMS
 - email
- Ensure seamless transitions (e.g., chatbot → human agent) with full context passed automatically.

8. Continuous Improvement Through AI Training

- Use generative AI to analyse logs and identify:
 - gaps in agent training
 - confusing steps in the call flow
 - reasons for escalations
- Automatically generate improved knowledge-base articles.

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Gemini:

Analysis of Innovative Bank's Current Use Case Issues

The current customer support system at Innovative Bank is suffering from classic scaling and personalization problems, leading to a negative customer experience.

Issue Category	Specific Flaws in the Current System	Impact on Customer Experience
Lack of Personalization	Scripted responses are inflexible and fail to address unique or complex customer queries. Agents are trained for consistency over individualized problem-solving.	Customers feel like a number, not a valued client. They are often asked to repeat information.
Inefficiency & Delays	Five broad inquiry categories lead to misrouting. Multiple transfers are required (Step 4 & Main Scenario) to resolve even simple issues (like a lost card).	Increased waiting time (up 20%) and longer call handling times. Customers are frustrated by the run-around.
Agent Limitations	Agents cannot resolve unique queries and must defer them for later analysis (Alternative Scenario). There is a reliance on queues and linear process flow.	Customers must wait up to 24 hours for a callback for issues that should be resolved immediately. This drives dissatisfaction and account closure.
Documentation Process	Documentation (Step 5) is done after the call, making real-time insight and	The system is slow to adapt to new problems, perpetuating old flaws.

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	immediate system updates impossible.	
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 **Generative AI for Customer Support Improvement**

Generative AI, especially a powerful LLM like Gemini or ChatGPT, can transform this outdated call-based process into a **Personalized, Expedient, and Proactive** customer experience platform.

Task 2: Creative Prompts for Improvement

The following set of prompts could be used to gather comprehensive ideas from a generative AI model:

1. **Initial Prompt (Establish Context & Goal):**
 "Analyse the following customer support process for 'Innovative Bank.' Identify key pain points for both customers and agents, focusing on the reliance on scripted responses, long wait times, and transfers.
Current Call Process: [Insert the 'Innovative Bank Call Process' Steps 1-5 here.]
Goal: Propose a completely revised, step-by-step customer support use case that incorporates generative AI (specifically a sophisticated LLM) to address these flaws, reduce agent load, and improve the customer satisfaction rate."
2. **Follow-up Prompt (Focus on Agent Empowerment):**
 "In the revised use case, how can the generative AI tool act as a '**Co-Pilot**' for the human agent, specifically for the Alternative Scenario (complex, unique issue requiring follow-up)? Detail the real-time AI capabilities that prevent the need for a 24-hour callback."
3. **Refinement Prompt (Focus on Efficiency and Personalization):**
 "Draft a new, optimized script for the initial customer interaction. The generative AI should handle **authentication** and **intent classification** simultaneously, replacing the five broad inquiry categories with dynamic, personalized routing based on the customer's account data and historical issues."

Revised Use Case with Generative AI

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Generative AI shifts the process from a linear, reactive, call-based system to a **dynamic, proactive, omnichannel experience**.

The Improved Innovative Bank Customer Support Use Case

Objective: Provide personalized, real-time, high-quality support across all channels, resolving over \$95\%\$ of inquiries on first contact (or automated resolution).

The New AI-Enhanced Support Process

Step	Action	Generative AI Role & Benefit
Step 1: Intake & Real-Time Context	Customer initiates contact via phone, chat, or app message.	The Generative AI Chatbot/IVR identifies the customer using voice or digital credentials . It instantly queries the customer's profile, recent transactions, and past support history.
Step 2: Dynamic Intent Classification	The customer states their need (e.g., "I lost my credit card").	The AI uses advanced LLM natural language understanding to classify the intent precisely (e.g., not just "Lost Card" but "Emergency Temporary Lock and Reorder"). No broad categories.
Step 3: Automated Resolution (Tier 0/1)	For \$80\%\$ of issues (e.g., balance check, lost card, payment dispute):	The AI instantly generates a unique, non-scripted solution . For a lost card, it executes the card lock/reorder API immediately, confirms via text, and provides an estimated delivery date. No queue, no human agent.
Step 4: AI Co-Pilot & Hand-off (Tier 2)	For unique/complex issues (e.g., loan application detail, complex fraud):	The AI automatically routes the call to the BEST-FIT Human Agent (not just the next available). The AI provides the

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		agent with a real-time "Briefing Card" containing: <i>Customer Profile, Full History, Transaction Details, and Suggested Next Steps/API commands.</i>
Step 5: Agent Empowerment	The Human Agent resolves the issue with AI assistance.	The AI acts as a "Co-Pilot," drafting responses, querying internal knowledge bases, and executing complex API calls based on the agent's voice command. No transfers or 24-hour callbacks. The AI can find solutions instantly.
Step 6: Proactive Learning & Documentation	Resolution is complete.	The AI automatically documents the unique resolution , tags it for \$N\$ (New) Knowledge, and updates the core LLM's knowledge base immediately. Real-time, comprehensive, and feeds back for continuous improvement.

Generative AI Benefits

- **Maximized Personalization:** Every response is context-aware and specific to the customer's history and current problem.
- **Reduced Wait Time:** Most inquiries (Tier 0/1) are resolved instantly by the AI, drastically shrinking the human agent queue.
- **Increased Agent Efficiency:** Agents handle more complex cases because the AI manages repetitive tasks and provides powerful real-time support, reducing call handle time and eliminating the need for internal transfers.
- **Improved First-Call Resolution:** The AI's access to all data and ability to execute actions means fewer issues are deferred for a 24-hour callback.

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